# Gambling Apocalypse Part I: Who Is Actually Smart Money?

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**Abstract**

In prediction markets, are historical winners merely fortunate, or do they possess superior forecasting ability? Can wallet trading behavior provide additional predictive information beyond market implied probabilities? This study investigates Smart Money in Polymarket by examining the persistence of forecasting skills, the predictive value of trading behavior, and changes in betting size.

Using approximately 570 million historical trades from January 2025 to March 2026, this study constructs a dataset covering 1.17 million wallets and 14,925 markets. Permutation tests and Empirical Bayes estimation are employed to examine forecasting skills, while Logistic Regression, LightGBM, MLP, Mean Pooling, and GRU models are developed to evaluate the predictive value of wallet behavior.

The results reveal statistically significant persistence in forecasting performance ($p \approx 0.007$), with stronger signals observed among experienced wallets. Empirical Bayes estimation improves the future forecasting performance of the selected top 10% wallets by approximately 44%. Further analysis shows that incorporating behavioral and historical trading features improves probability forecasts, while controlled experiments provide encouraging evidence of the additional information contained in historical trading behavior.

An intriguing finding concerns changes in betting size. Wallets tend to increase their stakes on outcomes already favored by the market. However, the largest additional forecasting advantage associated with larger bets occurs when market probabilities are between 40% and 60%. Within this interval, wallets placing unusually large bets outperform those maintaining similar stakes by 3.05 and 3.69 percentage points across two evaluation periods. These findings suggest that identifying Smart Money requires consideration of both historical forecasting performance and the circumstances under which wallets adjust their trading behavior.

## 1 Data and Research Design

### 1.1 Prediction Markets and Wallet Behavior

Polymarket is a blockchain based prediction market where participants trade contracts on the outcomes of future events. A typical market offers YES and NO contracts. For instance, in a market concerning whether a candidate will win an election, the price of a YES contract can be interpreted as the market's estimated probability of that outcome. Once the event is resolved, the contract corresponding to the correct outcome pays 1, while the alternative pays 0. Prediction markets aggregate the expectations of participants with different information, experience, and forecasting abilities. The resulting market probabilities provide a benchmark for evaluating individual forecasts. At the same time, differences in trading behavior may reveal information that has not been fully incorporated into market prices.

Blockchain transaction records provide an opportunity to examine these relationships at the wallet level. By tracing wallet addresses, it is possible to observe how the same wallet participates in different markets

over time. This study constructs wallet level features that describe historical forecasting performance, trading activity, betting size, and changes in trading behavior, allowing their predictive value to be investigated systematically.

## 1.2 Dataset Construction

The dataset covers Polymarket transactions from January 2025 to March 2026, comprising 569.54 million trade records. The original market registry contains 612,892 markets. By incorporating official historical market prices, identifying realized outcomes, and aligning market information with transaction timestamps, this study constructs a research dataset containing 14,925 markets and 1,169,132 wallets. The initial processing produces approximately 7.14 million observations indexed by wallet, market, and month. For predictive modeling, the first qualifying taker BUY made by each wallet in each market is selected. Historical features are then constructed using trading information available before the corresponding transaction. The resulting dataset for Behavioral Information Alpha contains 4,681,530 observations. The sample is divided into three chronological periods, as summarized in Table 1.

**Table 1:** Temporal division of the research dataset

| Dataset | Period | Observations |
|---|---|---|
| Training | Jan 2025 to Sep 2025 | 2,139,267 |
| Validation | Oct 2025 to Dec 2025 | 568,424 |
| Final Test | Jan 2026 to Mar 2026 | 1,973,839 |

This chronological division allows models to be developed using earlier observations and evaluated against subsequent market outcomes. It also provides a consistent basis for examining whether forecasting signals remain informative across different periods.

## 1.3 Market Benchmark and Evaluation

Historical market implied probabilities are used as the primary forecasting benchmark. These probabilities are obtained from completed three hour price intervals provided by Polymarket and aligned with individual transactions according to their timestamps. The main sample uses probability observations available within 12 hours before each trade. Two evaluation approaches are adopted to examine forecasting skills and behavioral information.

First, forecasting skill is evaluated by comparing the realized outcome of a wallet's prediction with the probability implied by the market before the transaction. The forecasting advantage of an individual trade is defined as

$$A = Z - q, \tag{1}$$

where $Z$ equals 1 if the outcome purchased by the wallet occurs and 0 otherwise, while $q$ denotes the market implied probability of that outcome.

For example, suppose a wallet purchases an outcome with an implied probability of 40%, and the outcome eventually occurs. The forecasting advantage of this observation is $1 - 0.4 = 0.6$. Aggregating these observations across markets provides a measure of wallet forecasting performance relative to market expectations.

Second, Behavioral Information Alpha is evaluated through probability forecasting. The Brier Score and Log Loss are employed to measure the difference between predicted probabilities and realized outcomes. A lower score indicates more accurate forecasts. Comparing models with different sets of behavioral features allows the incremental predictive contribution of trading information to be assessed. Each market receives equal weight during model evaluation, preventing markets with a larger number

of wallet observations from dominating the results. Statistical uncertainty is estimated using Bootstrap resampling clustered at the event level.

# 2 Skill vs. Luck: Can Past Winners Keep Winning?

## 2.1 How Persistent Is Forecasting Skill?

Wallets that perform exceptionally well in historical prediction markets may possess superior forecasting ability. However, favorable random outcomes can also produce impressive historical records, particularly when wallets participate in only a limited number of markets. To examine whether forecasting skill persists over time, this study compares the historical forecasting performance of individual wallets with their subsequent performance. Among 14,870 wallets that participate in at least five historical markets and five future markets, the correlation between historical and future forecasting scores is approximately 0.0234.

A permutation test is conducted to assess the statistical significance of this relationship. By controlling for differences in the number of forecasting opportunities available to individual wallets, the test constructs a reference distribution against which the observed correlation can be evaluated. The adjusted correlation is approximately 0.0227. Based on 1,000 permutations, the observed relationship is statistically significant, with a $p$ value of approximately 0.007. The results provide statistical evidence of persistent forecasting skill in prediction markets. Historical forecasting performance contains information about subsequent performance, indicating that a measurable component of forecasting ability persists across different market periods.

## 2.2 Experience Matters

The persistence of forecasting skill may depend on the amount of historical experience available for evaluating each wallet. A longer forecasting record provides more observations from which individual performance can be estimated, potentially allowing genuine skill to be distinguished more clearly from random variation. To investigate this relationship, wallets are grouped according to the number of historical markets in which they have made valid predictions. The future forecasting performance of historically successful and unsuccessful wallets is then compared within each experience group.

The results reveal a more pronounced forecasting advantage among experienced wallets. For the 1,367 wallets with at least 50 historical markets and 50 future markets, the top 10% based on historical forecasting performance achieve an average future forecasting score of approximately 0.0405. The corresponding score for the bottom 10% is approximately 0.0025, yielding a difference of 0.0380. The permutation test further confirms the statistical significance of this difference, with a $p$ value of approximately 0.001. These findings highlight the importance of experience in identifying Smart Money. Wallets with more extensive forecasting records provide stronger evidence about their forecasting ability, making persistent skill more distinguishable in historical transaction data.

## 2.3 Can Empirical Bayes Identify Better Winners?

The presence of persistent forecasting skill raises a practical question: how can wallets with superior forecasting ability be identified more accurately from historical observations? A conventional approach is to rank wallets according to their historical forecasting scores. However, extreme historical performance may reflect both genuine forecasting ability and favorable random outcomes. This phenomenon, commonly known as the Winner's Curse, is particularly relevant when individual wallets have limited forecasting histories. To improve wallet selection, this study employs Empirical Bayes estimation. The method adjusts observed forecasting scores according to the number of historical observations and their estimation

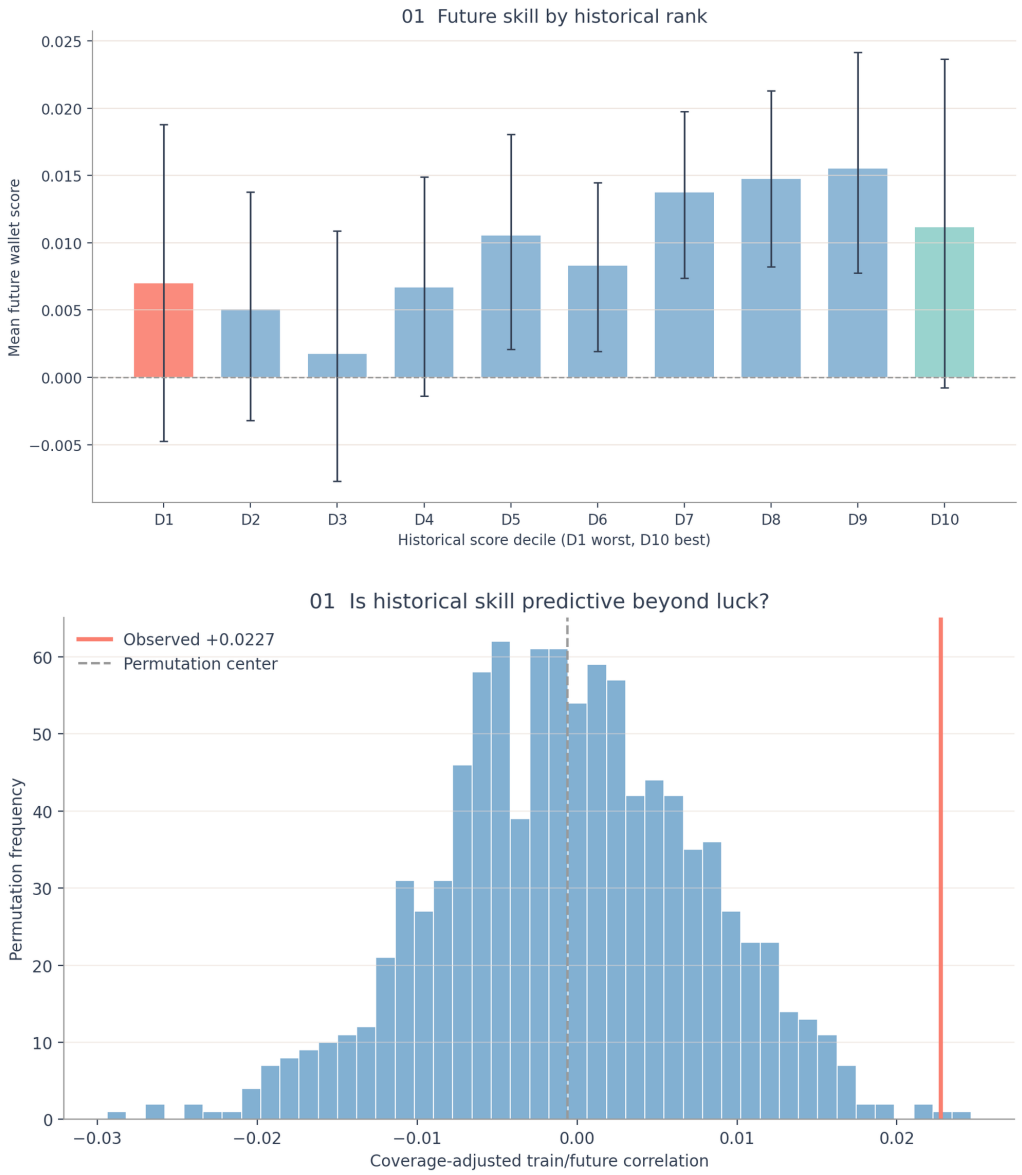



**Figure 1:** Historical skill deciles and coverage adjusted permutation test for 14,870 wallets with at least five historical and five future markets. Observed adjusted correlation: 0.0227; permutation $p = 0.007$ (seven day outcome availability specification).

uncertainty. Wallets with limited historical evidence receive greater adjustment toward the population average, while those with more extensive records retain a larger proportion of their individual performance estimates.

The results show that Empirical Bayes estimation substantially improves the identification of wallets with stronger future forecasting performance. Under the seven day outcome availability specification, the top 10% of wallets selected by conventional historical ranking achieve an average future forecasting score of approximately 0.0130. The corresponding score increases to 0.0188 under Empirical Bayes ranking, representing an improvement of approximately **44%**. Interestingly, only about half of the wallets selected by the two ranking methods overlap. The statistical adjustment therefore produces a substantially different group of potential Smart Money wallets. A similar improvement is observed under the 30 day specification, where Empirical Bayes ranking increases the average future forecasting score of the selected top 10% by approximately 79.5%.

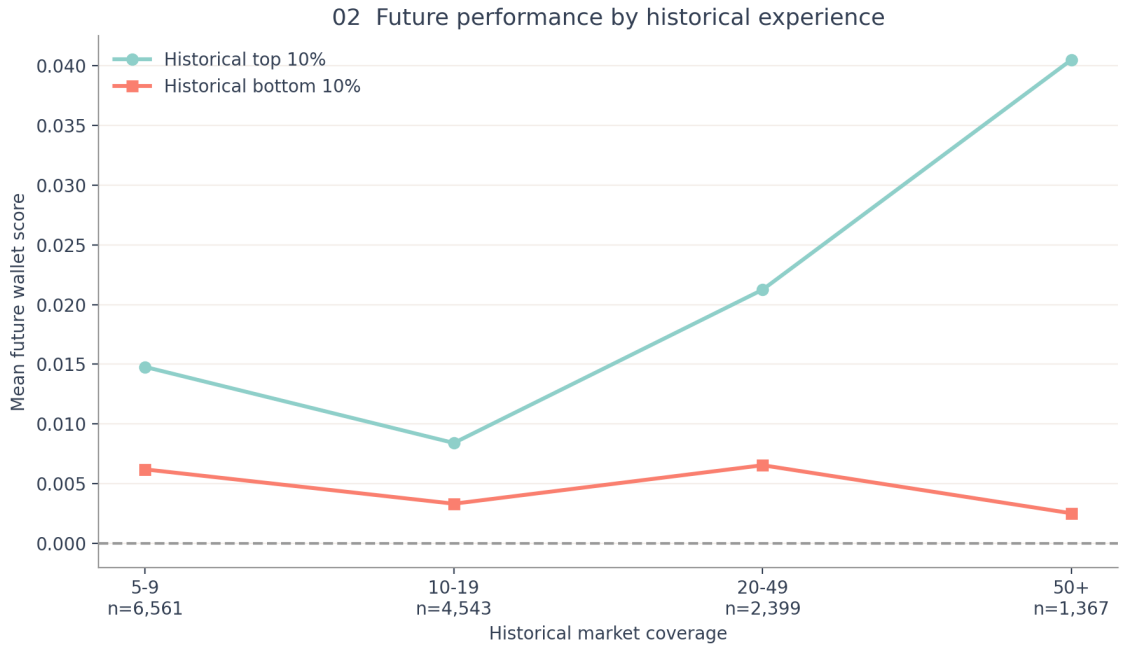



**Figure 2:** Future market adjusted forecasting score for wallets in the top and bottom historical deciles, grouped by historical market coverage. Wallet counts for each group are shown along the horizontal axis.

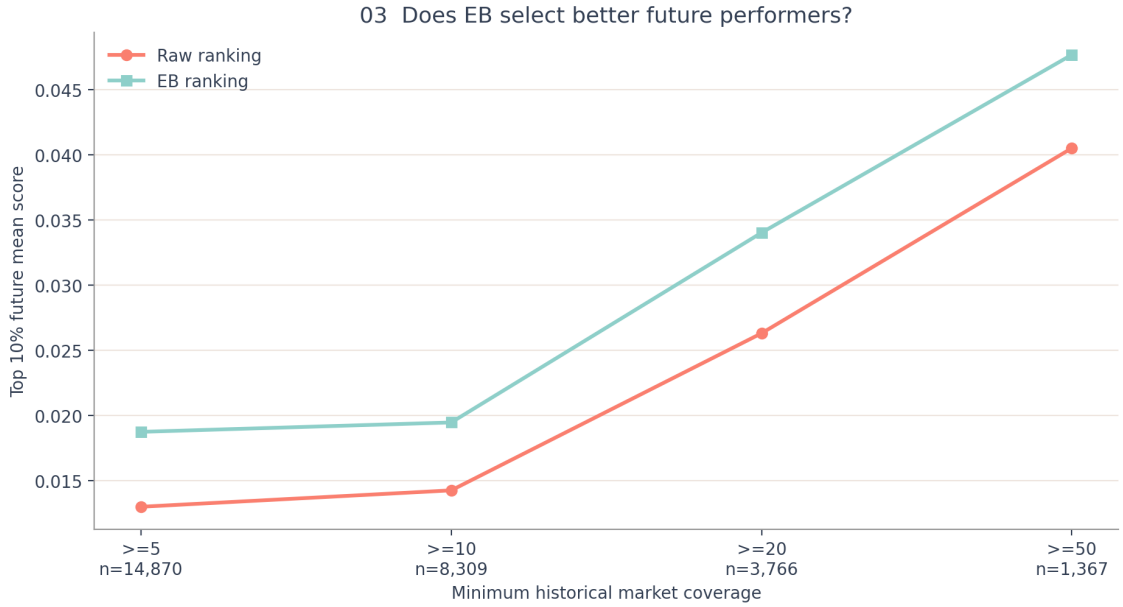



**Figure 3:** Future forecasting score of the selected top 10% wallets using raw and Empirical Bayes rankings across minimum historical market counts. Seven day outcome availability specification.

These findings demonstrate that the identification of Smart Money depends not only on the presence of forecasting skill but also on how that skill is estimated. By accounting for the uncertainty associated with historical performance, Empirical Bayes provides a more effective approach to selecting wallets with stronger future forecasting outcomes.

# 3 Behavioral Information Alpha: Does Wallet Behavior Add Predictive Information?

## 3.1 Beyond Market Probabilities

The previous analysis establishes that historical forecasting performance contains information about future wallet performance. A further question is whether the trading behavior of individual wallets provides additional predictive information beyond market probabilities. Consider two wallets purchasing the same outcome in the same market. Although both face the same market probability, their trading sizes, historical experience, recent activity, and previous forecasting performance may differ considerably. These behavioral differences may reflect distinct information or decision processes and consequently contribute to forecasting accuracy.

To examine this possibility, this study develops a series of predictive models with increasing amounts of information. The analysis begins with raw market probabilities, followed by probability calibration, anonymous trading features, and historical wallet characteristics. Logistic Regression and LightGBM are employed to evaluate the contribution of each information layer. Table 2 summarizes the results obtained from Logistic Regression during the final evaluation period.

**Table 2:** Forecasting performance with additional behavioral information

| Model | Brier Score |
|---|---|
| Raw Market Probability | 0.129032 |
| Market Calibration | 0.127967 |
| + Anonymous Behavior | 0.127915 |
| + Wallet History | **0.127858** |

The results show that probability calibration improves forecasting accuracy relative to raw market probabilities. Incorporating anonymous trading behavior further reduces prediction errors, while historical wallet characteristics provide an additional improvement. This pattern indicates that wallet behavior contains predictive information beyond the probabilities already reflected in the market. The improvements are relatively small, suggesting that much of the available information is already incorporated into market prices.

## 3.2 Learning from Historical Trading Behavior

Beyond the information contained in the current transaction, historical trading behavior may reveal how individual wallets make decisions over time. Conventional feature engineering typically represents trading history through summary statistics, including the number of previous trades, average trading size, and historical forecasting performance. However, wallets with similar summary statistics may exhibit substantially different trading patterns. For example, one wallet may consistently place small bets of similar size, while another may trade infrequently before occasionally making much larger bets. Such differences may contain useful information about trading behavior that is not fully captured by simple historical averages.

To investigate this possibility, this study constructs a sequence of up to 16 historical trades for each qualifying observation. The sequence contains information about the selected outcome, trade size, market probability, the age of historical price observations, and the time elapsed between trades. Three neural network structures are developed to evaluate different representations of historical behavior. The Static MLP uses current trading information and static wallet characteristics. Mean Pooling combines historical trading events into a representation of overall trading behavior. GRU further incorporates the temporal ordering of historical events.

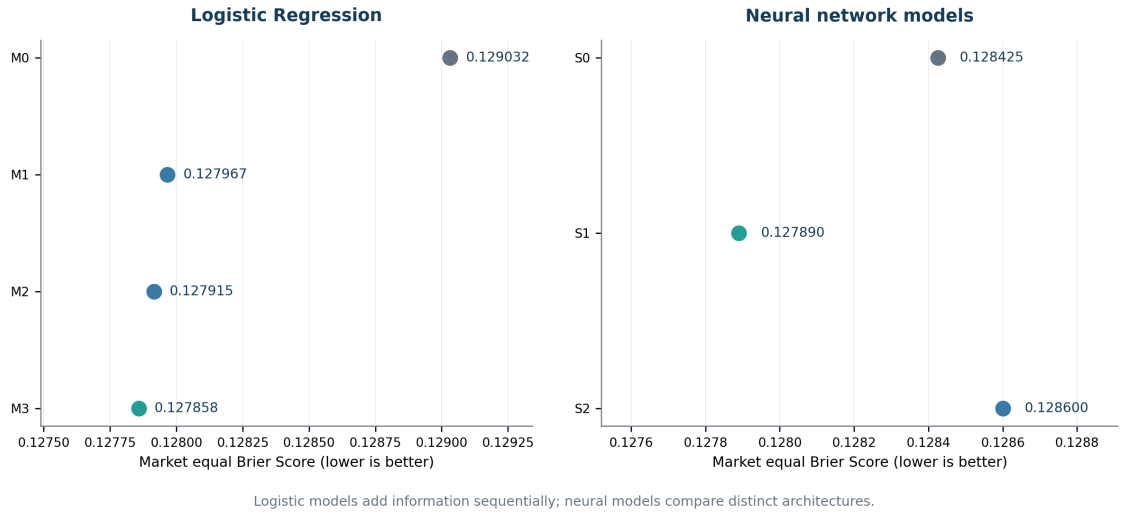



**Figure 4:** Market equal Brier Scores for the Logistic Regression model ladder (M0–M3) and the deep learning architectures (S0–S2). Scores are evaluated on the same 1,973,839 observations from Jan–Mar 2026; lower values indicate more accurate forecasts.

The results show that Mean Pooling achieves a Brier Score of 0.127890 during the final evaluation period, outperforming both the Static MLP, with a score of 0.128425, and GRU, with a score of 0.128600. These results suggest that a relatively simple representation of historical trading activity can capture useful predictive information. In this application, the additional complexity of GRU does not improve forecasting performance over Mean Pooling, indicating that the overall characteristics of historical behavior may be more informative than their precise temporal ordering.

### 3.3 Is Historical Information Really Useful?

The improvement achieved by historical behavior models raises a further question concerning the source of their predictive advantage. In particular, it is important to distinguish the contribution of historical trading information from improvements arising from differences in model architecture. To address this question, this study first conducts History Ablation and History Shuffle experiments. These experiments examine changes in forecasting performance when historical information is removed or when the content of historical trading sequences is randomized. The results indicate that historical trading sequences contain useful predictive information. Among wallets with available trading histories, models using the original historical content generally achieve better forecasting performance than those using randomized histories.

A further controlled experiment is developed to identify the contribution of historical event information more directly. Two models with identical network structures are compared. The first model, C0, retains historical trading counts and static wallet characteristics but excludes the detailed content of historical events. The second model, C1, incorporates the original historical trading events using the same architecture. Across three different random seeds, C1 consistently achieves a lower Brier Score than C0. The ensemble results further show that the Brier Score decreases from 0.127466 to 0.127282 when the original historical information is included, representing an improvement of approximately 0.000184. The corresponding 95% Bootstrap confidence interval ranges from −0.000057 to 0.000391. The positive improvement provides encouraging evidence of historical information value, although its statistical reliability requires further investigation.

These experiments provide further insight into the information contained in historical wallet behavior. In particular, the controlled comparison indicates that the content of previous transactions may contribute

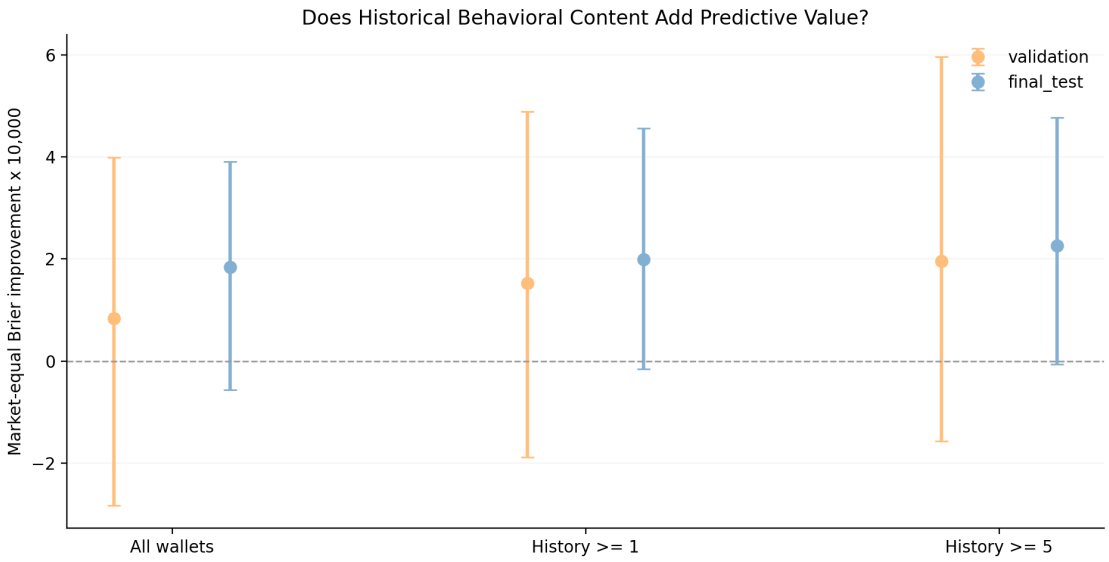



**Figure 5:** Incremental market equal Brier improvement from real historical event content (C1) compared with the same architecture without event content (C0). Event cluster Bootstrap 95% confidence intervals are shown for both evaluation periods.

to forecasting performance beyond the information already captured by trading frequency and static wallet characteristics.

### 3.4 When Does Historical Behavior Matter More?

The predictive contribution of historical trading information may also depend on the circumstances surrounding a wallet's current decision. To examine this relationship, observations are grouped according to the recency of historical trades, changes in betting size, changes in the probability of the selected outcome, and trading direction. The improvement associated with historical information is then evaluated across different behavioral conditions. The results reveal noticeable differences in predictive improvement across these groups.

Among observations with at least five historical trades, incorporating the original historical information reduces the Brier Score by approximately 0.000239 for wallets that have traded within the previous seven days. The corresponding improvement reaches approximately 0.000373 for observations classified as Higher Stake. These findings suggest that historical trading information may be particularly useful when wallets have been active recently or exhibit substantial changes in betting size.

Further comparisons within common markets show that some differences between behavioral groups become smaller after accounting for market characteristics. This motivates a more direct investigation of the relationship between betting behavior and actual forecasting performance. In particular, when a wallet suddenly increases its betting size, does this change correspond to a greater forecasting advantage relative to market expectations?

## 4 Betting Conviction: Do Bigger Bets Know Better?

### 4.1 When Wallets Suddenly Bet Bigger

Betting size is an important characteristic of trading behavior in prediction markets. A sudden increase in the amount committed to an outcome may reflect a change in how a wallet evaluates the corresponding event. To investigate this relationship, the current betting size of each wallet is compared with its own

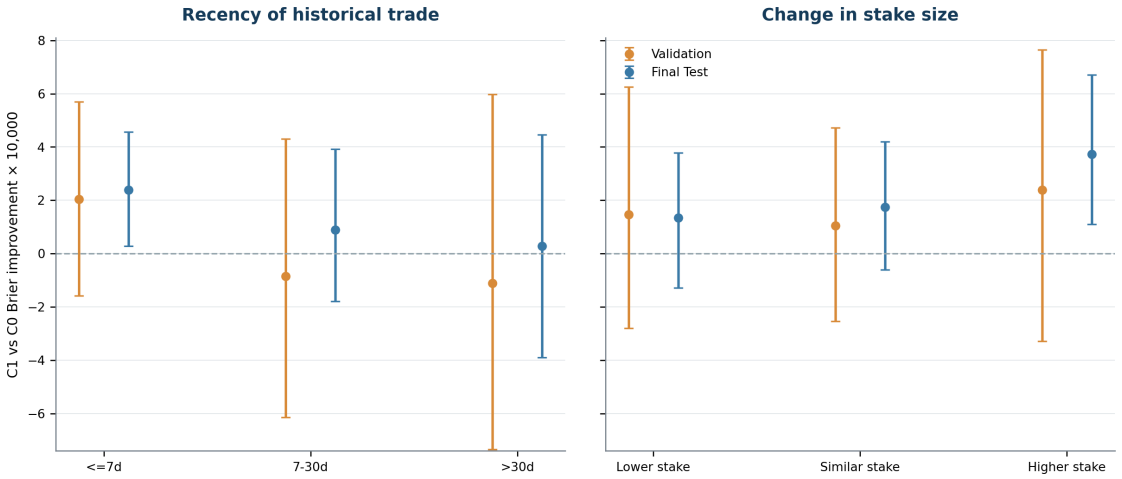



**Figure 6:** Incremental C1 versus C0 Brier improvement by historical trade recency and relative stake size, among observations with at least five historical trades. Error bars show event cluster Bootstrap 95% confidence intervals.

historical trading behavior. Trades are classified into three groups, namely Higher, Similar, and Lower Stake, according to whether the current stake is substantially above, close to, or below the wallet's historical level. This relative measure focuses on changes in individual trading behavior, allowing unusual betting activity to be identified among wallets with different typical transaction sizes.

The first finding reveals an interesting pattern: wallets tend to increase their stakes on outcomes already strongly favored by the market. During January to March 2026, 64.1% of trades in the Higher Stake group are associated with outcomes carrying market implied probabilities between 80% and 100%. The corresponding proportions are 26.7% for the Similar Stake group and 10.0% for the Lower Stake group. The results indicate that unusually large bets are frequently placed on outcomes that the market already considers highly likely to occur. This raises a further question concerning the information contained in larger bets. Are wallets making better predictions when they increase their stakes, or are they simply choosing outcomes with higher market probabilities?

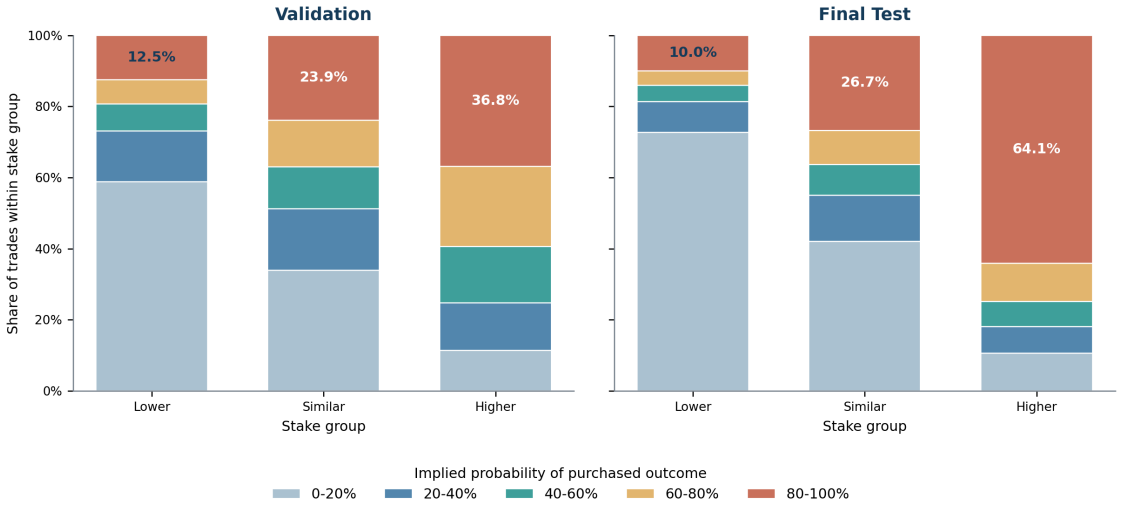



**Figure 7:** Composition of purchased outcome probabilities by relative stake group in Validation and Final Test, restricted to observations with at least five historical trades. The highlighted segments represent outcomes with market implied probabilities between 80% and 100%.

## 4.2 Does Betting Bigger Predict Better?

To investigate whether larger bets correspond to superior forecasting performance, this study evaluates the difference between realized outcomes and their corresponding market implied probabilities using the forecasting advantage measure defined in Section 1.3. Table 3 summarizes the forecasting performance of the three stake groups during the final evaluation period.

**Table 3:** Market adjusted forecasting performance across stake groups

| Stake Group | Implied Prob. | Win Rate | Advantage |
|---|---|---|---|
| Lower | 28.51% | 28.73% | +0.22 pp |
| Similar | 42.43% | 42.98% | +0.55 pp |
| Higher | 66.34% | 67.87% | **+1.53 pp** |

The Higher Stake group achieves both a higher realized success rate and a greater forecasting advantage relative to market expectations. To further examine the relationship, Higher Stake and Similar Stake observations are compared within markets in which both groups are present. Each market receives equal weight, allowing forecasting performance to be compared under more similar market conditions.

The results show that, during January to March 2026, the Higher Stake group achieves an additional forecasting advantage of approximately **1.09 percentage points** relative to the Similar Stake group. The corresponding 95% confidence interval ranges from 0.48 to 1.67 percentage points. Interestingly, the overall difference is considerably smaller in the preceding evaluation period. This variation suggests that the relationship between betting size and forecasting performance may depend on the characteristics of the underlying markets. To investigate this possibility, the next analysis examines how the forecasting advantage associated with larger bets varies with market implied probabilities.

## 4.3 The Conviction Paradox

If unusually large bets contain additional predictive information, under what market conditions is this information most valuable? The previous analysis shows that wallets frequently increase their stakes on outcomes already favored by the market. However, markets with probabilities close to 50% provide an interesting setting for investigating the informational value of betting behavior. In these markets, uncertainty remains relatively high, and individual forecasts may differ more substantially from prevailing market expectations. To explore this relationship, trades are divided into five groups according to the market implied probability of the selected outcome. The forecasting advantages of Higher Stake and Similar Stake observations are then compared within common markets and the same probability intervals. The results reveal a striking pattern, as summarized in Table 4.

**Table 4:** Additional forecasting advantage of Higher Stake relative to Similar Stake

| Market Probability | Validation | Final Test |
|---|---|---|
| 0–20% | -0.10 pp | -0.10 pp |
| 20–40% | +0.77 pp | +0.67 pp |
| **40–60%** | **+3.05 pp** | **+3.69 pp** |
| 60–80% | +1.32 pp | +0.39 pp |
| 80–100% | +0.10 pp | +0.05 pp |

The largest difference in forecasting advantage occurs when market probabilities lie between 40% and 60%. Within this interval, the Higher Stake group outperforms the Similar Stake group by 3.05 and 3.69 percentage points across the two evaluation periods. The corresponding 95% Bootstrap confidence

intervals range from 0.26 to 5.69 and from 2.15 to 5.28 percentage points, respectively. In contrast, when market implied probabilities are between 80% and 100%, the additional forecasting advantage associated with larger bets is close to zero in both periods.

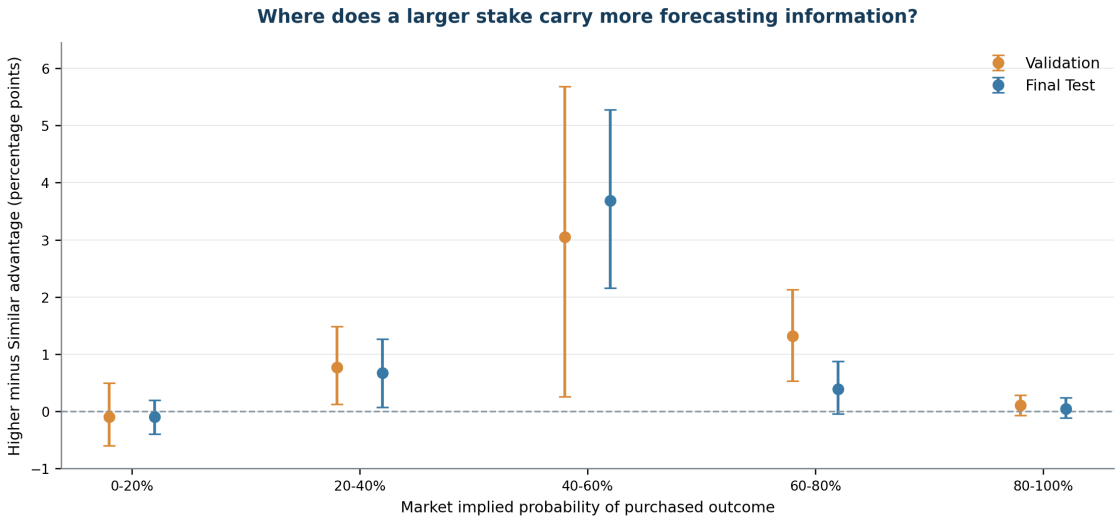



**Figure 8:** Higher Stake minus Similar Stake forecasting advantage within common markets and purchased outcome probability bins. Points are market equal differences in percentage points; bars show event cluster Bootstrap 95% confidence intervals.

These findings reveal an intriguing relationship between betting behavior and market uncertainty. **Wallets tend to increase their stakes most frequently on outcomes already favored by the market, whereas the additional forecasting advantage associated with larger bets is most pronounced when market outcomes remain uncertain.**

One possible explanation concerns the information already incorporated into market prices. When market expectations remain uncertain, changes in individual betting behavior may reveal information that has not yet been fully reflected in prevailing probabilities. As market expectations become more concentrated, the scope for additional predictive information from individual trading decisions may diminish. This interpretation provides a further perspective on Smart Money identification. Historical forecasting performance reveals which wallets have demonstrated predictive ability, while changes in betting behavior may indicate when their decisions contain additional information. The relationship between trading behavior and market uncertainty therefore offers an interesting direction for identifying informative wallet activity in prediction markets.

# 5 Discussion and Limitations

This study investigates Smart Money in Polymarket by examining forecasting skill, historical trading information, and changes in betting size. The results provide several insights into how predictive information is reflected in wallet behavior. First, historical forecasting performance exhibits statistically significant persistence, with more pronounced signals among experienced wallets. The Empirical Bayes analysis further demonstrates that accounting for estimation uncertainty improves the identification of historically successful wallets with stronger future performance. These findings indicate that historical forecasting records contain useful information, and that appropriate statistical estimation plays an important role in extracting persistent skill.

Second, wallet trading behavior contributes to forecasting accuracy beyond market probabilities in several predictive experiments. Historical trading information also demonstrates potential incremental value in controlled comparisons. The results suggest that wallet behavior can provide a complementary

source of information for probability forecasting, particularly when combined with market expectations. Third, the analysis of betting conviction reveals an interesting relationship between behavioral changes and market uncertainty. Larger bets are frequently placed on outcomes already favored by the market, while their additional forecasting advantage is most pronounced when market probabilities are close to 50%. This pattern suggests that the informational value of trading behavior may vary considerably across different market conditions.

Taken together, the findings suggest that Smart Money identification can benefit from incorporating both historical forecasting performance and dynamic trading behavior. Wallet experience, previous trading activity, changes in betting size, and market uncertainty provide complementary perspectives on the information contained in prediction market transactions.

Several aspects of the analysis warrant further investigation. Historical market probabilities may lag transaction times by up to 12 hours, while the availability of market outcomes is partly approximated. In addition, the research sample is determined by the availability of historical prices and resolved outcomes, and some comparisons depend on wallets remaining active in subsequent periods. Dependencies across related markets and events may also influence statistical inference. The betting conviction results are exploratory, as multiple behavioral comparisons were examined and the final evaluation period was inspected during the research process. Additional market periods and more precise transaction time probabilities would help assess the stability of the observed relationships.

Finally, the analysis focuses on predictive information rather than trading profitability. Translating these findings into practical trading strategies requires further consideration of transaction costs, market liquidity, execution prices, and trading opportunities. These issues provide a natural extension toward the investigation of Smart Money trading intelligence.

# 6 Concluding Remarks

This study examines the identification of Smart Money in Polymarket using historical forecasting performance and wallet trading behavior. By combining statistical inference, Empirical Bayes estimation, predictive modeling, and behavioral analysis, it investigates whether forecasting ability persists over time and whether individual wallet activity provides information beyond market probabilities. The results reveal statistically significant persistence in forecasting performance, particularly among experienced wallets. Empirical Bayes estimation further improves the selection of wallets with stronger subsequent forecasting performance. Historical trading behavior also exhibits potential incremental predictive value, demonstrating the usefulness of incorporating behavioral information into probability forecasting.

A particularly interesting finding concerns changes in betting size. Although wallets tend to increase their stakes on outcomes already favored by the market, the additional forecasting advantage associated with larger bets is most pronounced when market probabilities are close to 50%. This relationship highlights the potential importance of market uncertainty in understanding the informational value of wallet behavior. These findings suggest that identifying Smart Money involves more than evaluating historical forecasting records. The circumstances under which wallets change their trading behavior may also reveal valuable predictive information. The analysis provides a foundation for further research on information leadership, behavioral signals, and trading intelligence in prediction markets.